

Randomization tests

Test of independence

Participants in the study were 48 bank supervisors who identified as male and were attending a management institute at UNC in 1972. Each supervisor was asked to assume the role of personnel director of a bank.

- Each person given a file to judge whether the person in the file should be promoted
- The files were identical, except half of them indicated that the candidate was male, and the other half were indicated as female
- Files were randomly assigned to bank managers

Research question: Are individuals who identify their sex as female discriminated against in promotion decisions made by their managers who identify as male?

Step 1

H_0 :

H_A :

Step 2

```
discrimination |>  
  slice(1:3)
```

	sex	decision
1	male	promote
2	female	not promote
3	male	promote

sex	not promote	promote	total
female	10	14	24
male	3	21	24
total	13	35	48

Obtain some relevant/useful summary statistics:

$\alpha =$

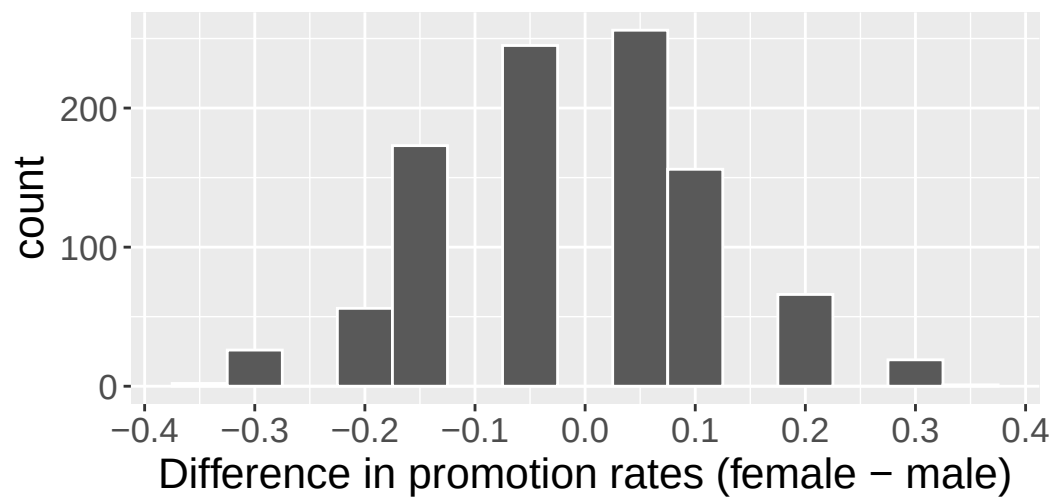
Step 3

Simulate under H_0 :

- Write down “promote” on _____ cards and “_____” on _____ cards.
- Then repeat the following B times:
 1. Shuffle the cards
 2. Deal out a stack of _____ to represent _____ candidates and _____ to represent _____ candidates.
 - 3.

```
1  set.seed(100)
2  n <- nrow(discrimination)
3  n_f <- sum(discrimination$sex == "female")
4  n_m <- sum(discrimination$sex == "male")
5  decisions <- discrimination$decision
6  B <- 1000
7  diff_props_null <- rep(NA, B)
8  for(b in 1:B){
9    shuffled <- sample(decisions, n)
10   rand_f <- shuffled[1:n_f]
11   rand_m <- shuffled[-c(1:n_f)]
12
13   p_f_sim <- mean(rand_f == "promote")
14   p_m_sim <- mean(rand_m == "promote")
15
16   diff_props_null[b] <- p_f_sim - p_m_sim
17 }
```

Null distribution



- p-value:

Step 4

- Decision:
- Conclusion:
- Possible error:

Test for difference in proportions

An experiment was conducted, consisting of two treatments on 90 patients who underwent CPR for a heart attack and subsequently went to the hospital. Each patient was randomly assigned to either:

- treatment group: received a blood thinner
- control group: did not receive a blood thinner

For each patient, the outcome recorded was whether they survived for at least 24 hours.

Research question: For patients who undergo CPR after a heart attack, does the blood thinner treatment have an effect on survival?

Step 1

H_0 :

H_A :

Step 2

```
cpr |>
  slice(1:3)
```

	group	outcome
1	treatment	died
2	control	died
3	control	survived

group	died	survived	total
control	39	11	50
treatment	26	14	40
total	65	25	90

Obtain some relevant/useful summary statistics:

$\alpha =$

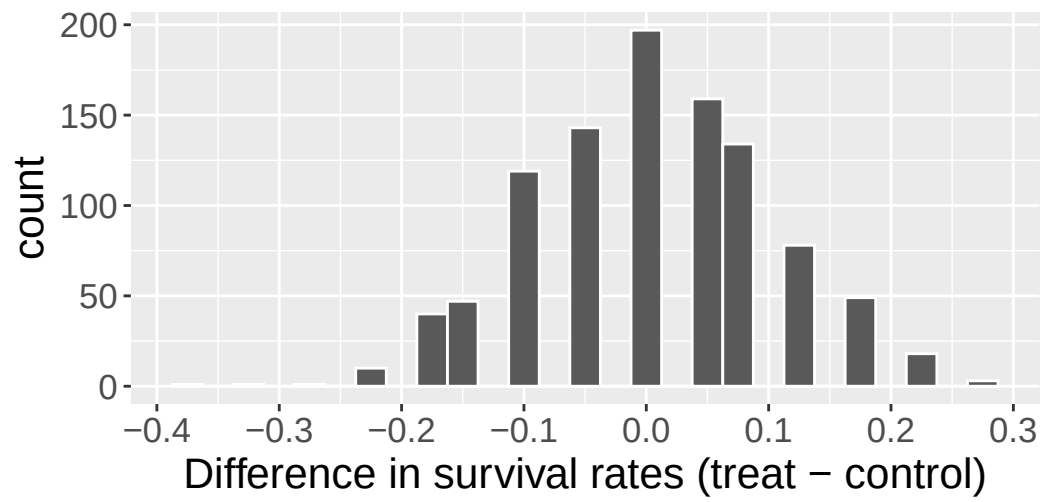
Step 3

Simulate under H_0 :

- Write down “_____” on _____ cards and “_____” on _____ cards.
- Then repeat the following B times:
 1. Shuffle the cards
 2. Deal out a stack of _____ to represent _____ candidates and _____ to represent _____ candidates.
 - 3.

```
1  set.seed(310)
2  n_t <- sum(cpr$group == "treatment")
3  n_c <- sum(cpr$group == "control")
4  cards <- cpr$outcome
5  B <- 1000
6  diff_props_null <- rep(NA , B)
7  for(b in 1:B){
8    shuffled <- sample(cards)
9    treat_sim <- shuffled[1:n_t]
10   control_sim <- shuffled[-c(1:n_t)]
11
12   p_t_sim <- mean(treat_sim == "survived")
13   p_c_sim <- mean(control_sim == "survived")
14
15   diff_props_null[b] <- p_t_sim - p_c_sim
16 }
```

Null distribution



- p-value:

Step 4

- Decision:
- Conclusion:
- Possible error: